## 3. Deep Critical Analysis: Six Foundation Papers

### 3.1 Christodoulou et al. (2024) - Science

CRAAP justification is hard to argue with - Science, 2024, five international RCTs via YODA, empirically direct rather than theoretical.

The numbers are stark: 0.72 balanced accuracy within training → 0.60 with cross-validation → 0.54 on independent trial data. Essentially coin-flip performance once the model leaves its training environment. Consistent across four different definitions of treatment success and both algorithm types (elastic net, random forest). 217 predictors, 1,513 patients. Difficult to dismiss as an artifact of small samples or weak design.

**Critical concerns:**

The representativeness problem is the biggest unacknowledged assumption. Schizophrenia is diagnostically heterogeneous, the drugs work on D2 receptors which is mechanistically distant from core pathophysiology - this might genuinely be one of the hardest clinical prediction problems that exists. The leap from "schizophrenia trials fail" to "all clinical ML fails" is a larger inferential jump than the paper defends. The finding is real and important but the scope claim is overextended.

The overfitting vs. transportability problem is unresolved and consequential. These two explanations require different fixes - overfitting needs better regularisation and more data; transportability failure needs causal methods, domain adaptation, invariant learning. The experimental design cannot separate them, so nobody reading this paper knows which problem they're actually dealing with.

The most glaring gap: the paper never tests anything from the distributional robustness toolkit. No importance weighting, no domain-adversarial training, no IRM. The conclusion that "models are fragile" is justified. The implicit message that the problem can't be solved is not - it's just untested.

Also worth flagging: "external" validation here means different pharmaceutical company trials, not different hospitals or routine care settings. Real clinical heterogeneity is much larger. The failure documented might actually be an underestimate of the real-world problem.

**Contribution to the gap:** Establishes empirically that the problem is real and serious. Stops at diagnosis - cannot say what a better approach would look like.

### 3.2 Whalen et al. (2022) - Nature Reviews Genetics

Strong CRAAP profile: 2022, Nature Reviews Genetics, interactive computational notebooks alongside the paper (rare for a review, genuinely useful for reproducibility). Selected as the most systematically organised account of technical failure modes in the corpus.

Five pitfalls: distributional differences, non-independent examples, hidden confounders, data leakage in preprocessing, class imbalance. Each demonstrated quantitatively - the dependent-examples pitfall can inflate AUPR by more than 0.5, leaky preprocessing can produce striking accuracy on datasets with no real biological signal. These are reproducible demonstrations, not just assertions.

**Critical concerns:**

The paper is built on the IID assumption - fix IID violations, model generalises. That framing is useful but incomplete. A model can satisfy all IID requirements and still fail at deployment because the deployment population differs from training in ways no preprocessing checklist captures. The Christodoulou failure, as far as can be determined, is not caused by leakage or class imbalance - it's caused by statistical relationships not transferring across clinical environments. Whalen et al.'s framework doesn't address that.

The confounding treatment is the most specific weakness. Confounding is listed as one of five equivalent pitfalls, but causal confounding is qualitatively different from the others. Including confounders as features or using PCA to capture unmeasured structure are reasonable steps but don't guarantee causal stability. A model learning age-outcome associations in a 40-year-old-heavy cohort has captured that population's specific age-outcome relationship. When deployed on a different age distribution, that relationship may not hold - not because of preprocessing error, but because the relationship is age-dependent in ways the model can't account for.

The "unified remedy" - make training-test split mirror real-world deployment - is correct in principle and underspecified in practice. Clinical deployment happens across many sites, over years, in changing contexts. Often the deployment scenario isn't known in advance.

**Contribution to the gap:** The pre-flight checklist. Limitation is that passing the checklist doesn't guarantee generalisation once the model is deployed. The gap between "passed the checklist" and "will work in deployment" is where the research problem sits.

### 3.3 Wörheide et al. (2021) - GigaScience

Moderate CRAAP profile - 2021, open-access, well-regarded but not top-tier, field has moved quickly since publication. Selected because it's one of the few papers addressing dataset shift specifically in clinical ML biomarkers, and because one central finding - that a widely-used corrective practice actively makes performance worse - is important enough that all biomedical ML researchers should know it.

Theoretical framework: covariate shift (features change, outcome relationship stable) vs. prior probability shift (prevalence changes, manifestation stable). These require different corrections - conflating them means applying the wrong fix.

Experiment: predicting smoking status from UK Biobank with intentional age-distribution gap. Three approaches compared: unchanged model, importance weighting (reweighting training data to match test age profile), deconfounding (regressing age out of features before training).

**Key finding:** Regressing out age makes performance worse in every configuration tested, including configurations with no shift at all. Not marginally worse - consistently and meaningfully worse. This directly contradicts common practice in biomedical ML. Evidence is clear and the claim is directly supported.

**Critical concerns:**

The evidence-to-recommendation gap is the main problem. One type of shift, one dataset, one shift direction (younger to older). Real clinical shifts are multivariate - age, ethnicity, comorbidity patterns, equipment generations, clinician behaviour, coding practices, regional disease epidemiology all changing simultaneously and only partially observable. The theoretical framework handles this in principle; the experimental evidence for whether corrections work covers only the univariate case.

The importance weighting recommendation has a boundary condition acknowledged but not investigated: it only works if every subgroup in the target population appears at least somewhat in training data. The paper doesn't show how performance degrades as coverage drops from complete to partial - exactly the information a practitioner would need.

The DRO (distributionally robust optimisation) recommendation is presented as forward-looking strategy but is not tested anywhere in the paper. Recommending an untested method is claiming beyond the evidence.

**Contribution to the gap:** Provides tools for correcting known, measured shifts. Has nothing to say about shift that occurs after deployment in real time when the nature of the shift isn't known in advance. That operational question - detecting drift, characterising it, deciding what to do - is entirely unaddressed.

### 3.4 Emerson et al. (2023) - Nature Machine Intelligence

Strong CRAAP profile: 2023, Nature Machine Intelligence, constructive rather than diagnostic. Selected as the most technically complete constructive proposal in the corpus - the paper that comes closest to saying "here is what to do instead."

Framework: DAGs to explicitly map cause-and-effect relationships before building any model. Identifying which relationships are genuinely causal (stable across environments) vs. spurious correlations (will evaporate when context changes). Applied to immune receptor repertoire (AIRR) diagnostics - using T-cell and B-cell receptor sequence patterns to diagnose disease.

Three simulation experiments: (1) confounder distribution shift → balanced error rate nearly doubles; (2) selection bias in training → model looks excellent during development, performs at chance on unbiased data; (3) batch effects in AIRR data substantially harder to correct than in gene expression data, and correction risks removing real biological signal.

**Critical concerns:**

The simulation methodology is both the greatest strength and the most significant constraint. No one knows the true causal rules governing real immune receptor data, so synthetic datasets with defined ground truth are the only rigorous option. But that means experiments test what happens when the model's assumptions are violated in specifically programmed ways - not in the chaotic, multivariate, partially-unobservable ways real clinical data generates. Results tell how the method performs against its own idealised scenarios, not against reality.

The causal vs. anticausal prediction distinction is underexplored relative to its importance. Causal prediction: inferring effect from cause (predicting immune response after vaccination). Anticausal prediction: inferring cause from effects (using receptor patterns to diagnose autoimmune disease). Anticausal models are shown to be less stable when input feature distribution changes - they're working backwards against the causal direction of the data-generating process. Most diagnostic tasks in clinical medicine are anticausal. If anticausal prediction is systematically less stable, that's not a niche finding about immune receptors - it's a fundamental vulnerability of the whole diagnostic ML enterprise. The paper identifies this but doesn't pursue it to its logical conclusion.

The causal graph specification problem is the Achilles heel. Every missing edge is an assumption that two variables aren't causally related. In high-dimensional biological systems, the empirical basis for ruling out specific relationships is often weak. The paper is transparent about this limitation but doesn't address the critical question: what happens when the graph is partially wrong? Do you get a model better than a naive associative approach, or worse? All simulations use correctly-specified graphs, so this question is unanswerable from the paper.

**Contribution to the gap:** Closest to a constructive solution in the corpus. The open limitation - practitioners need to specify a causal graph before the framework helps, and often can't do so reliably - points toward a specific tractable contribution: developing methods for learning or approximating causal structure from data when prior knowledge is incomplete.

### 3.5 Springer and Doering (2021) - Briefings in Bioinformatics

Moderate CRAAP profile - 2021, solid but not exceptional authority. Relevance is indirect (immunological sequence prediction, not clinical ML broadly). Included because it's the only paper in the corpus seriously confronting what might be called the boundary condition of generalisation: what happens when the observations being generalised to belong to categories that simply didn't exist in training data? Not new patients with familiar characteristics - genuinely novel biological territory.

ImRex: treats TCR-epitope binding prediction as image classification. Amino acid sequences represented as 2D matrix of pairwise physicochemical properties, CNN applied to learn patterns from that matrix.

Most important contribution is the evaluation distinction: seen-epitope (model tested on epitopes encountered during training, even with different receptor partners) vs. unseen-epitope (epitopes never in training data at all). Most prior work reported only the first metric. On the second, performance drops dramatically - often to chance. The decoy experiment is the key: several published models performing well at receptor-epitope binding prediction performed equally well when epitopes were replaced with random nonsense sequences. Those models weren't learning interaction patterns - they were memorising receptor sequence characteristics alone.

**Critical concerns:**

The architectural justification is absent. CNNs work on images because of spatial locality and translation invariance - nearby pixels are related, and the same feature at different positions is conceptually the same. Neither property obviously holds for an amino acid interaction matrix. Positions that matter for binding aren't necessarily adjacent in the matrix representation, and there's no biochemical reason to think a hydrophobicity mismatch at position 3 is equivalent to one at position 7. The method outperforms baseline, but without theoretical grounding, it's unclear whether this would hold on different sequence families or is specific to the epitope distribution in this database.

The negative data problem is more serious than acknowledged. Non-binding pairs have to be generated artificially (databases only contain confirmed binding pairs). The method - pairing a known TCR with a randomly selected epitope it's not known to bind - assumes TCRs only ever bind their catalogued partners. But TCR cross-reactivity is well-established. Some "negative" examples generated by shuffling are probably genuine binding pairs mislabelled as non-binding. The paper doesn't estimate how large this contamination is or what effect it would have on the model's learned decision boundary.

The headline finding that the method can extrapolate to unseen epitopes has a substantial condition attached: extrapolation only works for epitopes within approximately one amino acid substitution of something in training data. That's interpolation at the very edge of the training distribution, not generalisation in any meaningful sense.

**Contribution to the gap:** Maps the theoretical limit of the problem. When target observations belong to genuinely new categories, standard approaches fail completely and best alternatives only work at the boundary of what was seen before. Any generalisation research either needs to address this boundary condition with direct evidence, or explicitly define its scope to exclude it with justification.

**3.6 Collins et al. (2024) - TRIPOD+AI, BMJ**

Highest currency in the set (2024, most recently updated global reporting standard). Exceptional authority: BMJ, EQUATOR registered, Delphi process with 200 international participants from research, clinical practice, regulation, and patient advocacy. Selected not as the most technically sophisticated paper but as the normative boundary within which all other research must operate.

27-item checklist for reporting clinical prediction model development and evaluation, traditional regression or modern ML. Three pillars: discrimination, calibration, clinical utility. Advances over 2015 predecessor: fairness embedded across 12 items, open science practices (pre-registration, code sharing) explicitly mandated, patient/public involvement required.

**Critical concerns:**

The core limitation: TRIPOD+AI is a transparency instrument, not a quality instrument. A model can comply with all 27 items and still be deeply unreliable in deployment. The five schizophrenia trial models in Christodoulou et al. could have been reported in perfect TRIPOD+AI compliance and still averaged 0.54 balanced accuracy on independent data. The checklist makes problems visible - it doesn't fix the underlying data-generating problem.

The Delphi process has a structural bias toward conservatism. Items that are technically optimal but unfamiliar, controversial, or complex tend to fail the 70% consensus threshold not because they're wrong but because consensus is harder to achieve for ideas challenging existing practice. The fairness items that made it in are primarily about reporting subgroup performance - necessary but not sufficient to address structural causes of minority group under-representation in training data.

The LLM/foundation model gap is acknowledged and real. Foundation models are entering clinical practice rapidly and TRIPOD+AI can't be applied to them in current form.

The most directly useful criticism for research purposes: Item 26 requires discussing generalisation limitations, but has no defined standard of fulfilment. A single sentence saying "this model may not generalise to other populations" satisfies Item 26 in the same way as a detailed analysis of measured distribution shift, anticipated deployment contexts, and evidence-based transportability estimates. That ambiguity means the item can't be meaningfully enforced or evaluated.

**Contribution to the gap:** Defines the reporting floor any new work on prediction model robustness must meet. The constructive opportunity it creates is precise: specifying what an adequate characterisation of generalisation limitations should actually look like - with measurable criteria, not just a requirement to acknowledge the problem exists. Filling that specification gap is a direct and practical contribution.

**<u>Cross-paper observations</u>**

The two papers doing the heaviest lifting for the research gap argument are Christodoulou (establishes the problem empirically) and Emerson (comes closest to a constructive solution but leaves the causal graph specification problem open). The gap between them between knowing models fail and having a practical framework for predicting and preventing that failure - is the space the research is trying to occupy.

The overfitting vs. transportability distinction raised in the Christodoulou critique recurs implicitly across multiple papers and never gets resolved. That unresolved question is probably the most important single diagnostic problem in the whole literature.

The anticausal prediction point in Emerson et al. feels underweighted throughout the review. If most clinical diagnostic tasks are anticausal and anticausal models are systematically less stable, that's a structural problem not just for immune receptor diagnostics but for clinical ML broadly.